

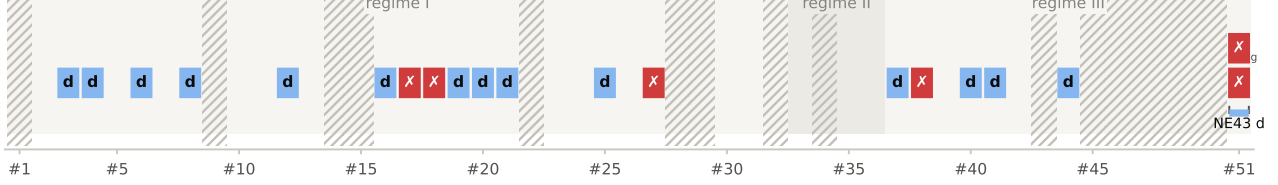
H72: Trap aging is input starvation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 70% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Trap aging is carried by the context's self-share, not by missing input; room chatter adds a weak hold that a mention cancels.

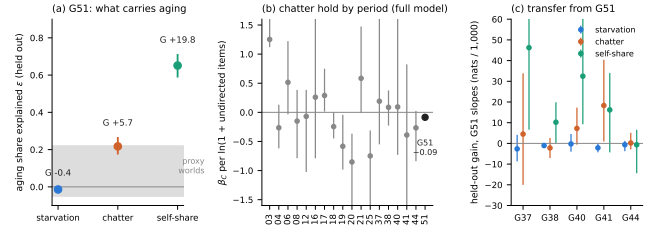
level: descriptive · pipeline checks 86% · updated 2026-10-05



Goal periods tested (6 failed, 15 descriptive). ✔ supported ✗ failed ⚡ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The claim: an idle agent's escape slows with trap age only because no new input has reached it. A wake is a call after an idle call; there the agent acts or idles again. Round 1 refuted the claim in-sample: the age slope survives every starvation clock (G51 $-0.48 \rightarrow -0.53$), and recent input *lowers* escape. Round 2 asks which input or context variable carries the aging, on the same 20,237 G51 wakes, scored on held-out days. Directed-input starvation explains none of it ($\epsilon = -0.01$). Undirected chatter explains 22%. The context's self-share (H16's own-call urn) explains 65% [59, 71], against $\leq 22\%$ in synthetic worlds where aging is intrinsic or frailty. Its coefficient is the urn's +1 (1.09), and its slope transfers from G51 to G38. Chatter adds a small, separate hold ($\beta_C = -0.085$) that a directed read cancels. The reserved data are not used.



Observables. (a) #51: the share of the aging clocks' held-out information that each mechanism makes redundant, with its own held-out gain (nats per 1,000 wakes). Grey band: the most a stand-in for wake index or per-trap frailty gives in synthetic worlds. (b) Chatter-hold slope per period, full model; only G51 is powered. (c) G51's slopes scored on the other regime-III periods: self-share helps in G37, G38 and G40; starvation never helps.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 09 (exogenous input events) and model 02 (kinetic escape) with an urn on the context. At wake g the agent acts (a run of ≥ 3 active calls) or idles. Clocks: trap age a and wake index k . Three mechanisms compete on one base (agent effects, pause, time of day, others' activity, reads at the wake, reset step, trap kind at start). S : minutes since the last directed read. C : undirected items read at the 5 calls before the wake. F : own idle calls as a share of own calls in the context segment. Traps that start with a consolidation are dropped.

$$\begin{aligned} \text{logit } P_g &= \alpha_i + z_g \cdot \delta + \beta_a \ln a_g + \beta_k \ln k_g \\ &+ \beta_S \ln s_g + \beta_C \ln(1 + U_g) + \beta_F \ln(1 - f_g) \\ \epsilon(M) &= 1 - \frac{\mathcal{L}_{B+M+A} - \mathcal{L}_{B+M}}{\mathcal{L}_{B+A} - \mathcal{L}_B} \quad (\text{held out}) \\ \text{rates race: } \beta_{\ln \lambda_u} &= -1; \quad \text{urn: } \beta_F = 1 \end{aligned}$$

H72-trap-aging-input-starvation · generated 2026-10-04 21:25 by infra/summaries/build_summaries.py · card hypotheses/H72-trap-aging-input-starvation/README.md · ratings: H72 round-2 agent (suggested v1; coordinator writes v2)

RESULTS VS PREDICTIONS

prediction (round 2)	observed (#51)	verdict
chatter dose holds (full model)	$\beta_C = -0.085$ $[-0.140, -0.045]$	holds
separable (< 30% change)	$-0.130 \rightarrow -0.085$ (34%)	fails
first wakes; agent + day FE	$-0.076; -0.090$	holds
a mention cancels the hold	$+0.14$ $[+0.06, +0.21]$	holds
rates race (slope -1)	-0.094 $[-0.148, -0.041]$	holds
replication (none above 0)	below 0: 4/18; above 0: G03, G17	fails
starvation explains aging	no $\varepsilon_S = -0.01$	holds
self-share explains most beyond proxy worlds (0.22)	$\varepsilon_F = 0.65$ vs $\varepsilon_C = 0.22$	holds
aging left over	$\varepsilon_{SCF} = 0.69$ $[0.64, 0.75]$	holds
self-share transfers to G38	$+10$ $[+0, +20]$ nats/1,000	holds

GOAL PERIODS IN DETAIL

G03 d descriptive
G04 d descriptive
G06 d descriptive
G08 d descriptive
G12 d descriptive
G16 d descriptive
G17 $\times$ failed
G18 $\times$ failed
G19 d descriptive
G20 d descriptive

+11 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

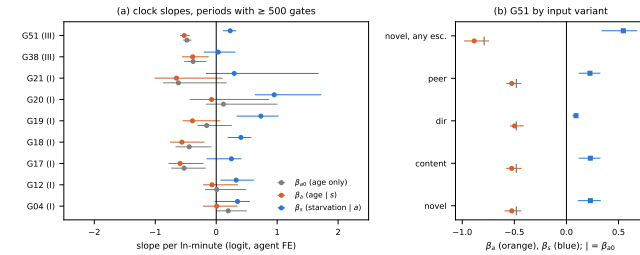
Only G51 is powered for the chatter hold; G03 and G17 have CIs above 0, and other periods are descriptive. A third of the naive dose slope is shared with self-share and starvation. The in-flight placebo is non-zero (+0.29) but leaves β_C unchanged; escaping calls may have longer model latency. Self-share and wake index correlate (-0.69); synthetic proxy worlds bound the overlap at 0.22. 31% of the clocks' information remains unexplained. The transfer fit was fixed post hoc (step halving).

ROUND 2

Confirm ε_F and the urn coefficient on reserved #51 days and #45–#50 with a re-frozen confirm script. Find what carries the last third of aging (memory, task state). Explain the in-flight term.

- **H72-R1** (done in round 2, 2026-10-05).
- **H72-R2** (done in round 2, 2026-10-05).
- **H72-R3** (open; needs re-freezing on round-2 statistics).

H72-trap-aging-input-starvation · generated 2026-10-04 21:25 by infra/summaries/build_summaries.py · card hypotheses/H72-trap-aging-input-starvation/README.md · ratings: H72 round-2 agent (suggested v1; coordinator writes v2) (page 2)



Second figure. Round 1. (a) Age slope without s (grey), with s (orange) and the starvation slope (blue), periods with ≥ 500 wakes, day-block 95% CIs: starvation absorbs no aging. (b) G51 under four input definitions and the glance outcome: the age slope never moves.